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Jing Wei

jingwei@pku.edu.cn

Peking University

Yulong Fan

Peking University

Zhanqing Li

University of Maryland, College Park <https://orcid.org/0000-0001-6737-382X>

Lin Sun

Shandong University of Science and Technology

Yuan Wang

Stanford University <https://orcid.org/0000-0001-6657-8401>

Hongbin Yu

Chuanfeng Zhao

Peking University <https://orcid.org/0000-0002-5196-3996>

Jing Li

Peking University <https://orcid.org/0000-0002-0540-0412>

Huizheng Che

State Key Laboratory of Severe Weather Meteorological Science and Technology (LaSW) & Key Laboratory of Atmospheric Chemistry of CMA, Chinese Academy of Meteorological Sciences, Beijing, China <https://orcid.org/0000-0002-9458-3387>

Zhengqiang Li

Aerospace Information Research Institute, Chinese Academy of Sciences

Lin Chen

China Meteorological Administration

Alexei Lyapustin

NASA Goddard Space Flight Center

Oleg Dubovik

Université de Lille <https://orcid.org/0000-0003-3482-6460>

Yiran Peng

Tsinghua University <https://orcid.org/0000-0002-3686-5458>

Xinghua Qiu

Peking University

Mei Zheng

Peking University

Jian Gao

State Key Laboratory of Environmental Criteria and Risk Assessment, Chinese Research Academy of Environmental Sciences <https://orcid.org/0000-0001-8605-8454>

Tong Zhu

Peking University <https://orcid.org/0000-0002-2752-7924>

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GEO-Ring observations reveal substantially overestimated global land aerosol cooling

Yulong Fan^{1,2}, Jing Wei^{1*}, Zhanqing Li^{3*}, Lin Sun², Yuan Wang⁴, Hongbin Yu⁵, Chuangfeng Zhao⁶, Jing Li⁶, Huizheng Che⁷, Zhengqiang Li⁸, Lin Chen⁹, Alexei Lyapustin¹⁰, Oleg Dubovik¹¹, Yiran Peng¹², Xinghua Qiu¹, Meng Zheng¹, Jian Gao¹³, Tong Zhu¹

1. MEEKL-AERM, College of Environmental Sciences and Engineering, Institute of Tibetan Plateau, and Center for Environment and Health, Peking University, Beijing, China
2. College of Geodesy and Geomatics, Shandong University of Science and Technology, Qingdao, China
3. Department of Atmospheric and Oceanic Science, Earth System Science Interdisciplinary Center, University of Maryland, College Park, USA
4. Department of Earth System Science, Stanford University, Stanford, CA, USA
5. Earth Science Division, NASA Goddard Space Flight Center, Greenbelt, MD, USA
6. Department of Atmospheric and Oceanic Sciences, School of Physics, Peking University, Beijing, China
7. State Key Laboratory of Severe Weather Meteorological Science and Technology and Key Laboratory of Atmospheric Chemistry of CMA, Chinese Academy of Meteorological Sciences, Beijing, China
8. Key Laboratory of Remote Sensing and Digital Earth & Key Laboratory of Satellite Remote Sensing of Ministry of Ecology and Environment, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China
9. National Satellite Meteorological Center, China Meteorological Administration, Beijing, China
10. Laboratory for Atmospheres, NASA Goddard Space Flight Center, Greenbelt, MD, USA
11. Laboratoire d'Optique Atmosphérique, Université de Lille, CNRS, Lille, France
12. Department of Earth System Science, Ministry of Education Key Laboratory for Earth System Modeling, Institute for Global Change Studies, Tsinghua University, Beijing, China
13. State Key Laboratory of Environmental Criteria and Risk Assessment, Chinese Research Academy of Environmental Sciences, Beijing, China

*Corresponding authors: zhanqing@umd.edu; jingwei@pku.edu.cn

Abstract

The direct aerosol radiative effect (DARE) remains one of the largest uncertainties in Earth's climate system, partly because global estimates rely on polar-orbiting satellites with limited daily sampling. Here, we combine observations from eight geostationary satellites (the GEO-Ring constellation) with an integrated Transformer and transfer learning framework to retrieve global, hourly aerosol optical depth (AOD) at 550 nm and 2 km resolution over land. The resulting dataset agrees well with independent random-sample-based cross-validation ($CV-R^2 = 0.86$; $RMSE = 0.085$) and enables robust characterization of diurnal AOD variability. We find pronounced diurnal cycles, with a mean amplitude of $46\% \pm 29\%$ relative to a global mean land AOD of 0.17 ± 0.13 during 2021–2023, alongside strong regional contrasts. Instantaneous polar-orbiting observations yield systematically higher aerosol loadings by 7–48% across 60–80% of global land areas relative to GEO-Ring estimates, reflecting unresolved diurnal variability and retrieval differences. These discrepancies lead to a 35–79% overestimation of global annual mean DARE over land (-3.8 W m^{-2}), implying an inflated estimate of aerosol-induced cooling and consequently a greater challenge for mitigating global warming. Our results highlight the importance of resolving diurnal aerosol variability for robust quantification of aerosol radiative effects and their climate impacts.

Introduction

Atmospheric aerosols play a critical role in Earth's energy balance and cloud-precipitation processes by scattering and absorbing radiation, as well as acting as cloud condensation or ice nuclei¹⁻³. Variations in aerosol abundance and properties can modify radiative fluxes and cloud microphysics³, thereby influencing the occurrence and intensity of extreme weather events such as heatwaves⁴, heavy precipitation⁵, droughts⁶, and severe haze episodes⁷. Accurately constraining aerosol distributions and their radiative impacts is therefore essential for improving climate projections, attributing extreme events, and informing air-quality and sustainability policies.

Aerosols exert the direct radiative effect (DARE), which can produce net cooling (overwhelming) or warming (exceptional circumstances)⁸ depending on their composition, size, mixing state⁹, vertical distribution, and surface properties (e.g., albedo). Global assessments generally indicate a net cooling, with top-of-atmosphere DARE ranging from -1.6 to -7 W m⁻², based on ground-based observations, chemical transport models, and satellite data¹⁰⁻¹⁶. Despite decades of studies, DARE remains one of the most uncertain components of the climate system, largely due to incomplete knowledge of aerosol spatiotemporal variability, particularly over land, where emissions, transport, and removal processes vary strongly across regions and timescales^{11,13,17,18}. Climate and chemistry–climate models typically capture synoptic-scale aerosol variability but substantially under-sampled diurnal variations¹⁹⁻²¹. In addition, short-lived extreme events such as wildfires and dust storms can generate rapid aerosol enhancements that are difficult to resolve with sparse observations²².

Satellite remote sensing has revolutionized global aerosol monitoring. Polar-orbiting sensors, such as the Moderate-resolution Imaging Spectroradiometer (MODIS), provide long-term, near-global coverage²³. These observations underpin most global DARE estimates, either through radiative transfer calculations driven by satellite-derived aerosol optical depth (AOD)^{15,16,24,25} or by direct analysis of broadband radiation measurements from the Clouds and the Earth's Radiant Energy System (CERES)^{13,14,26}. However, polar-orbiting satellites observe each location only once or twice per day at mid-to-low latitudes, fundamentally limiting their ability to capture diurnal aerosol cycles and rapidly evolving episodic events. Consequently, instantaneous satellite observations are often averaged to represent daily means²⁷; however, global analysis using long-term observations

has revealed pronounced diurnal variations in AOD, indicating that daily mean AOD is inherently inaccurate when represented by limited overpasses of polar-orbiting satellites²⁸. This can introduce systematic biases in aerosol loading and radiative forcing estimates²⁹.

Many geostationary (GEO) satellites are currently in operation, providing near-continuous global coverage, including the Geostationary Operational Environmental Satellite-R series (GOES-R) over the Americas, Himawari Advanced Himawari Imager (AHI) over Asia–Oceania, China’s Fengyun-4 over Asia, and the Meteorological Satellite series (Meteosat) over Europe, enabling high-frequency aerosol observations at 10–15 min intervals. These GEO satellites offer a unique opportunity to overcome the temporal limitations of polar-orbiting platforms, but their limited spatial coverage, heterogeneous sensor characteristics, and the lack of a unified retrieval framework have hindered the production of consistent, global hourly AOD products^{30,31}. Machine-learning approaches have steadily improved aerosol retrievals from both polar-orbiting and geostationary platforms³²⁻³⁴. However, these models remain strongly dependent on ground-based measurements often reduces robustness in regions with sparse station coverage³⁵⁻³⁷, constraining their utility for global radiative assessments.

Here, we extend our previously developed integrated deep learning (Transformer) and transfer-learning framework (AeroTrans), originally designed for Meteosat Second Generation (MSG-1)³⁸, by integrating GEO-Ring observations from eight geostationary satellites to retrieve global, hourly and consistent AOD products at 550 nm over land at 2 km resolution. The model is pre-trained using simulated aerosol reanalysis fields and then fine-tuned with ground-based observations, which improves inter-sensor consistency and spatial generalization while reducing sensitivity to uneven station distributions³⁸. The resulting dataset enables direct characterization of global diurnal aerosol cycles and their spatial heterogeneity, as well as quantification of daily aerosol loadings and their impacts on daily mean DARE over land. We further use this dataset to assess how limited temporal sampling by polar-orbiting satellites biases estimates of aerosol loading and radiative forcing.

Results

Diurnal variations in global aerosol loadings

Using a constellation of eight geostationary satellites (GEO-Ring), including GOES-18, GOES-17 and GOES-16 over the Americas, MSG-1 and MSG-2 over Europe, Africa, and West Asia, and FY-4A and Himawari-8/9 over Asia and Oceania, we achieved, *for the first time*, global hourly retrievals of AOD (550 nm) over land for the period 2021–2023 using the refined AeroTrans-GEO model (Fig. 1a). This framework employs a time-sequence Transformer that ingests multispectral GEO-Ring observations (0.4–2.2 μm) along with auxiliary variables. The model is pre-trained with MERRA-2 AOD reanalysis and fine-tuned using AERONET and SONET AOD ground-based measurements (Fig. 1b). It demonstrates strong retrieval skill ($R^2 = 0.87$) and robust generalization across independent sample-based, temporal, and spatial ten-fold cross-validations ($\text{CV-}R^2 = 0.86, 0.85, \text{ and } 0.79$; Fig. 1c), with only moderate degradation at longer temporal and coarser spatial scales (Text S1). It also provides improved temporal representation and reduced uncertainties relative to estimates based on limited temporal sampling from polar-orbiting satellites ($\sim 10:30$ and $13:30$ local time), as evaluated against ground-based measurements (Fig. S4).

Global hourly AOD exhibits pronounced diurnal variability, ranging from a minimum of 0.11 ± 0.04 to a maximum of 0.19 ± 0.13 (00:00–23:00 UTC), highlighting strong sub-daily dynamics across regions and aerosol regimes. In arid and dust-dominated regions, such as Northwest Africa, the Middle East, and northwestern China, AOD consistently peaks in the early morning (06:00–08:00 local time), consistent with dust uplift and boundary-layer evolution (Fig. S5). Diurnal amplitudes typically range from 13–26% (25th–75th percentiles), but intensify to over 40% during extreme dust events, indicating rapid aerosol loading changes occurring within a few hours (Fig. S6).

Diurnal variability is even stronger in South Asia and East Asia, where mean amplitude exceeds 50% and exhibits distinct temporal structures, peaking in the early to mid-morning (07:00–10:00 local time). Northern India exhibits a largely monotonic daytime evolution, characterized by elevated AOD at sunrise, a mid-morning maximum ($\sim 10:00$ local time), and a gradual decline toward evening (Fig. S7a), driven primarily by human activities, boundary-layer growth, humidity, and stagnation meteorological conditions³⁹. In contrast, eastern China displays a clear bimodal cycle, with morning (07:00–08:00 local time) and evening (17:00–18:00 local time) peaks (Fig.

S7b), aligned with traffic-related emission timing⁴⁰. During severe haze episodes, diurnal amplitudes can exceed 60%, highlighting abrupt aerosol enhancements that occur on sub-daily timescales (Fig. S8).

Regions affected by biomass burning, including western North America, central Africa, Southeast Asia, and the Siberian Plain, also show strong diurnal variability, with amplitudes commonly exceeding 40% and increasing to ~60% during wildfire events, reflecting episodic emission pulses (Fig. S9). Even relatively clean regions such as Europe and Oceania exhibit measurable diurnal variability (~20%), highlighting the sensitivity of aerosol loading to short-term atmospheric and emission processes. Such substantial hourly aerosol variability is largely missed by polar-orbiting satellites with limited daytime coverage.

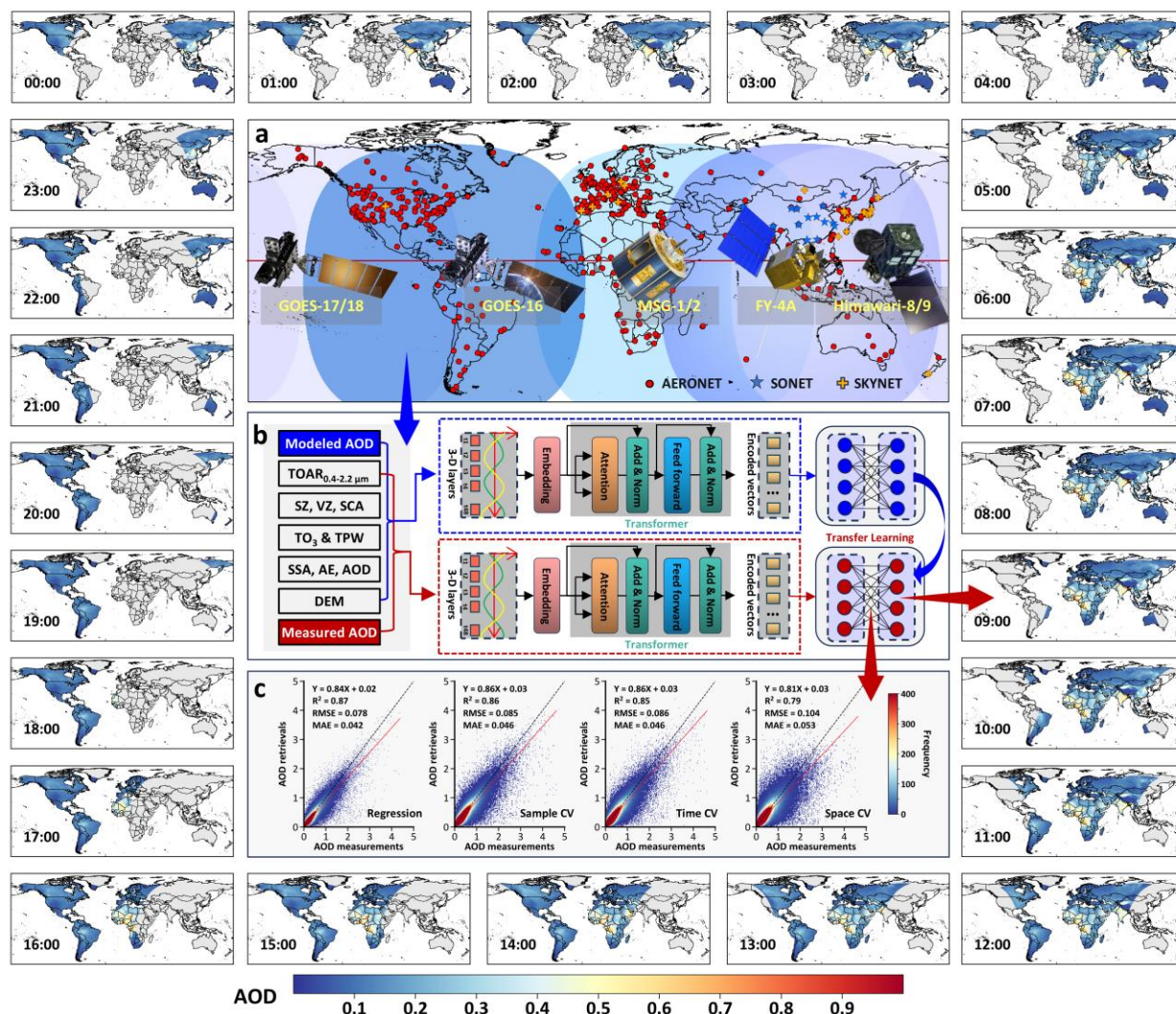


Fig. 1 | First global hourly AOD: dataset generation, validation, and diurnal variations.

(a) Observations from a constellation of eight geostationary satellites (GEO-Ring). (b) Workflow of the AeroTrans-GEO deep-learning framework with transfer learning. (c) Model validation, including regression performance and 10-fold cross-validation across sample, temporal (hourly), and spatial (site) dimensions. Surrounding panels show hourly clear-sky AOD (550 nm) at 2-km resolution over global land for 2021–2023 (00:00–23:00 UTC).

Pinpointing global aerosol loadings by GEO-Ring observations

Although GEO-Ring and polar-orbiting MODIS observations show strong agreement in large-scale AOD patterns, this apparent consistency conceals substantial regional and local discrepancies in aerosol magnitude, variability, and spatial coverage (Fig. 2). Positive differences are widespread across roughly 62% of global land area. In high-latitude regions ($> 60^\circ \text{ N}$), these differences can reach 80%, coinciding with large daily AOD variability (standard deviation $> 30\%$; Fig.2d) and limited retrieval coverage (up to $\sim 50\%$; Fig.2f), reflecting challenges from complex terrain and

frequent cloud and snow cover⁴¹. Over western North America, differences even reach 60%, driven by pronounced diurnal variability (amplitudes > 100%) and partial spatial coverage (Fig. 2f). In South Asia, MODIS often yields higher AOD than GEO-Ring, with differences commonly exceeding 20%, driven by episodic biomass-burning and urban pollution peaks⁴². GEO-Ring captures hourly fluctuations, smoothing transient extremes in daily aggregates, whereas MODIS snapshots may coincide with peak events, particularly during episodic phenomena such as wildfires and dust storms. Similar patterns are observed over the Amazon Basin, southern Africa, Southeast Asia, and Australia, where intense but short-lived aerosol plumes, combined with GEO-Ring's substantially improved retrieval coverage (> 80%), produce apparent MODIS-high values. Over Europe, positive differences are smaller (< 10%), reflecting moderate diurnal variability and occasional cloud-induced coverage gaps (Fig. 2c).

In contrast, within major aerosol source regions, such as North African and Middle Eastern dust hotspots and parts of Central Asia, GEO-Ring and MODIS diverge markedly, with daily AOD negative differences exceeding 30% on average and local discrepancies frequently surpassing 60% (Fig. 2c). GEO-Ring captures transient high-AOD events that polar-orbiting satellites, limited to fixed overpass times, often miss. Over the Tibetan Plateau, MODIS shows substantially lower AOD than GEO-Ring (differences exceeding 100%), likely reflecting complex terrain, frequent cloud and snow cover, and limitations of passive retrievals over bright surfaces. In contrast, GEO-Ring provides improved spatial coverage due to more frequent daytime observations, although uncertainties remain. Similar negative biases are evident over the eastern United States and central to southern South America, where daily differences exceeding 20% coincide with high variability, reflecting pronounced subregional heterogeneity.

The GEO-Ring-derived global daily AOD over land (0.17 ± 0.12) is, on average, ~7% lower than that from the Terra/Aqua MODIS MAIAC product (0.18 ± 0.11), with a relative standard deviation of 23%, highlighting pronounced regional and temporal heterogeneity. These differences exhibit a clear latitudinal pattern—overestimation at high latitudes and underestimation at low latitudes—primarily driven by the limited temporal sampling of sub-daily aerosol variability by LEO satellites, as evidenced by the similar spatial patterns between GEO-Ring and GEO-Ring-sampled (LEO-like overpass at 10:30 and 13:30 local time; Fig. S10). The discrepancies are especially pronounced

during December–January, when MODIS yields AOD values that are $33\% \pm 81\%$ higher than those from GEO-Ring (Fig. S11). Such differences primarily reflect strong diurnal variability, averaging $\sim 46\%$ globally (consistent with 48% from ground observations; Fig. S12) and reaching localized peaks exceeding 180%. Furthermore, GEO-Ring substantially enhances retrieval availability, achieving coverage across $\sim 99\%$ of global land. Together, its higher temporal sampling, more complete spatial coverage, and methodological differences enable GEO-Ring to more effectively capture daily and annual aerosol variability, including transient events often missed by polar-orbiting sensors.

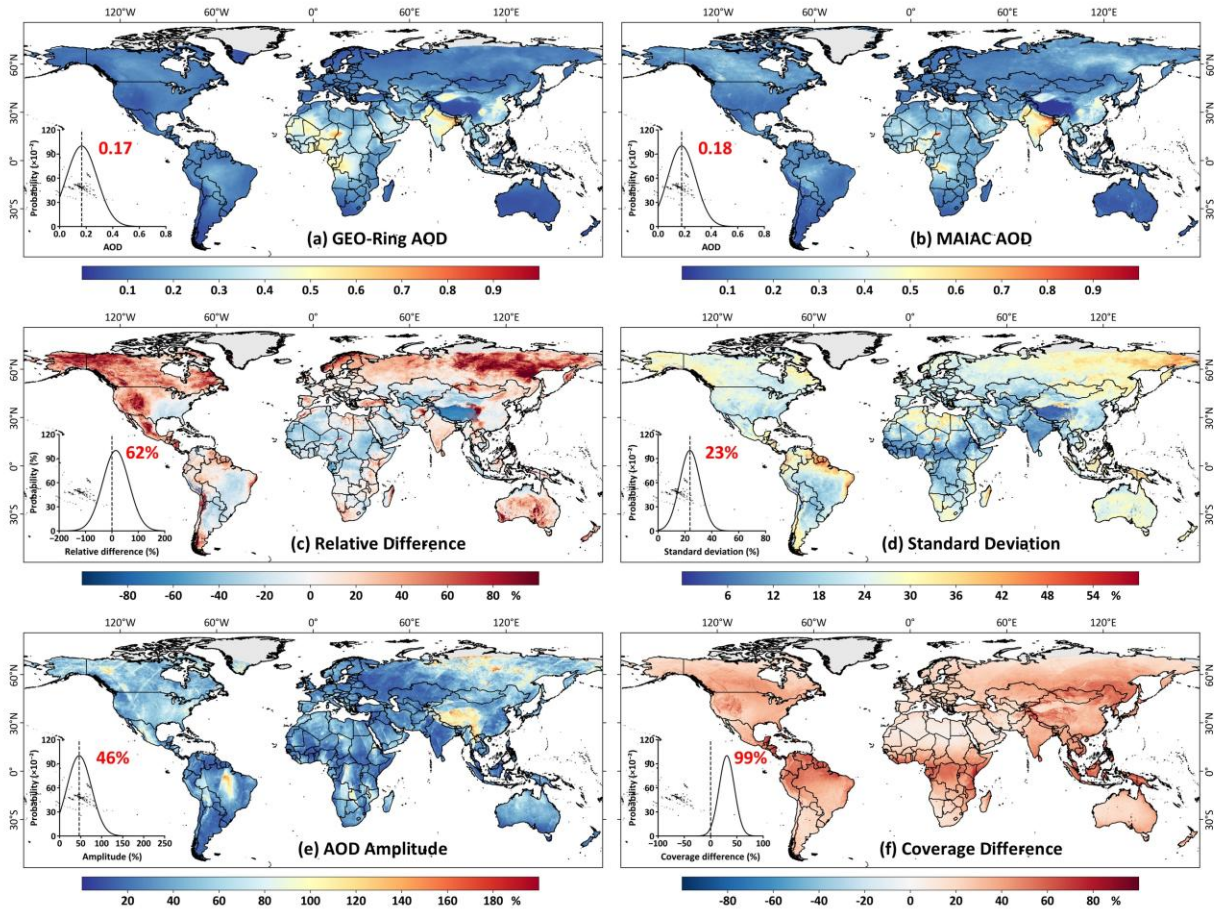


Fig. 2 | Discrepancies in global AOD from GEO-Ring and polar-orbiting observations.

Spatial distributions of annual AOD (2021–2023) derived from (a) GEO-Ring observations using the AeroTrans-GEO model and (b) Terra/Aqua MODIS observations using the MAIAC algorithm. (c) Relative difference (%), calculated as $(\text{MODIS} - \text{GEO-Ring}) / \text{GEO-Ring}$. (e) Diurnal amplitude of AOD, defined as $(\text{maximum} - \text{minimum}) / \text{daily mean}$ from GEO-Ring retrievals. (f) Difference in spatial coverage between GEO-Ring and MODIS daily AOD. Insets in panels (a–f) show frequency histograms. Red numbers indicate mean values in panels (a) and (b–e), and the percentage of pixels with positive differences in panels (c) and (f).

Pinpointing global aerosol direct radiative effects by GEO-Ring observations

Despite the fundamentally different sampling strategies, GEO-Ring and polar-orbiting MODIS observations yield broadly consistent large-scale spatial patterns of annual DARE under clear-sky conditions (Fig. 3a-b). Both datasets indicate widespread negative forcing over 91% and 94% of global land, respectively, with the strongest cooling signals ($< -10 \text{ W m}^{-2}$) occurring over densely polluted regions in South and East Asia, as well as fire-prone regions in central Africa. In contrast, positive forcing is primarily confined to bright-surface regions, particularly across North Africa and the Middle East. This overall consistency in the sign and geographic distribution of DARE demonstrates that the first-order controls on aerosol radiative forcing are robustly captured by both observing systems at climatological scales.

However, we find a systematic tendency for MODIS to yield higher annual DARE than GEO-Ring across approximately 26% of global land areas (Fig. 3c). These discrepancies are most pronounced in regions such as Central Asia and western China, where MODIS-derived DARE often exceeds GEO-Ring estimates by more than 40%. This bias likely stems from MODIS's polar-orbiting sampling strategy, which may fail to capture short-lived aerosol events and peak diurnal activity, as well as from differences in retrieval algorithms. Pronounced positive DARE differences are also observed across tropical biomass-burning regions, including South America (34%) and Central Africa (14%). This positive bias is likely driven by the combined effects of persistent fire activity and frequent cloud cover⁴³. The eastern United States exhibits relatively modest positive differences, associated with transient aerosol plumes and moderate diurnal variability. In contrast, the hourly observations from GEO-Ring more effectively capture the full diurnal cycle of aerosols, thereby minimizing the influence of short-lived extremes on estimates of daily mean DARE.

Negative differences are more spatially extensive, with the largest values often exceeding -80% and concentrated in high-latitude regions, primarily reflecting substantially higher availability of successful AOD retrievals from GEO-Ring compared with polar-orbiting observations (Fig. 2f). This pattern is also evident in dust- and fire-affected regions and is associated with large day-to-day variability, highlighting GEO-Ring's sensitivity to sub-daily aerosol dynamics. Notable examples include northwestern North America (-229%) and northwestern Africa (-53%), both of

which exhibit pronounced negative biases accompanied by extremely high variability in daily DARE differences, with standard deviations frequently exceeding 180% (Fig. 3d). In these regions, particularly during wildfire events, GEO-Ring effectively captures the rapid diurnal evolution and peak radiative forcing of smoke plumes, dynamics that are systematically missed by polar-orbiting observations, which encounter more cloud cover at that time and may misclassify heavy smoke as clouds. In anthropogenic pollution hotspots such as India (−18%), the negative differences are smaller but remain spatially coherent, suggesting that sub-daily sampling is still important even in regions dominated by anthropogenic aerosols.

The absolute difference highlights widespread and spatially heterogeneous discrepancies between the two observing systems, with a global mean of 2.2 W m^{-2} over land (Fig. 3e). Large differences are concentrated in regions characterized by episodic but intense aerosol events, coinciding with areas of strong diurnal aerosol variability. In these regions, twice-daily polar-orbiting observations are insufficient to capture rapid sub-daily changes in aerosol loading and radiative forcing. Overall, we pinpoint the global land DARE of -3.8 W m^{-2} ($60^\circ \text{ S} - 70^\circ \text{ N}$) based on GEO-Ring observations. In comparison, the Terra/Aqua MODIS MAIAC AOD product yields -5.3 W m^{-2} , indicating a ~39% more negative annual mean DARE and thus an overestimation of aerosol cooling due to unresolved sub-daily variability. These results demonstrate that sampling frequency can introduce substantial uncertainty into long-term DARE estimates, particularly under extreme aerosol conditions such as wildfires, dust storms, and severe haze (Figs. S13–S15).

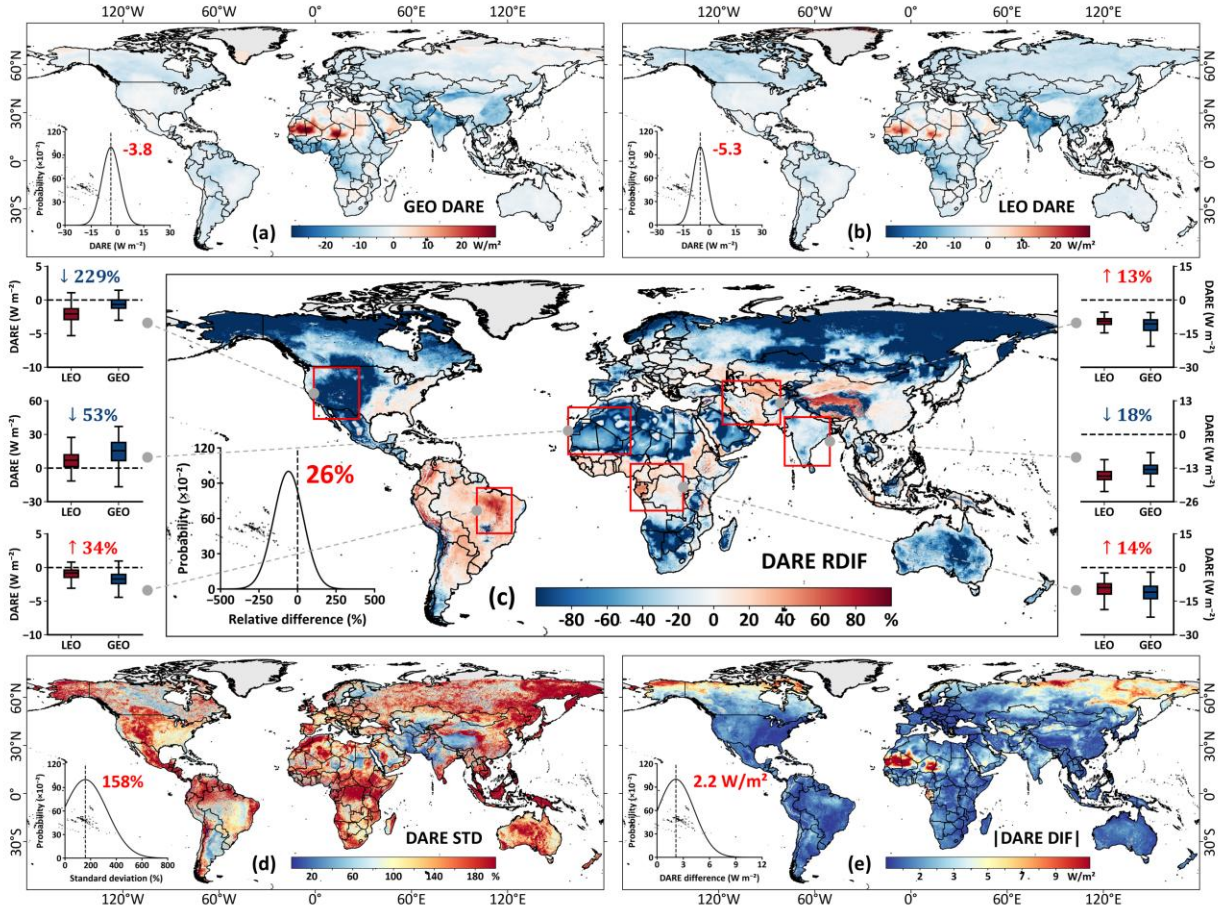


Fig. 3 | Discrepancies in global DARE from GEO-Ring and polar-orbiting observations.

Spatial distribution of the annual clear-sky direct aerosol radiative effect (DARE) derived from (a) hourly GEO-Ring observations using the AeroTrans-GEO model, and (b) twice-daily Terra/Aqua MODIS observations using the MAIAC algorithm during 2021–2023. Panels (c) and (d) show the relative difference (%) between GEO-Ring- and MODIS-derived annual DARE, and the daily standard deviation of these differences, respectively. Panel (e) shows the absolute difference between the two DARE estimates. Insets in (a–e) display frequency histograms. The surrounding boxplots summarize the annual DARE differences for selected regions: northwestern North America, South America, the western Sahara Desert, central Africa, central Asia, and India. Red numbers indicate mean values in (a–e), the differences in the surrounding subplots, and the percentage of pixels showing positive differences in (c).

Discussions

Notable differences in global land AOD loadings emerge across satellite retrieval products and platforms (Fig. 4). Similar to MAIAC, other algorithms exhibit consistent spatial patterns in AOD differences relative to GEO-Ring, with 60–80% of global land showing higher AOD than GEO-Ring (Fig. 4a-d). Among MODIS products, Terra/Aqua Deep Blue (DB) yields the highest AOD values, with a global annual mean AOD of $\sim 0.25 \pm 0.15$, approximately 48% higher than GEO-

Ring (Fig. 4e). Terra/Aqua Dark Target (DT) also produces higher AOD ($\sim 31\%$ above GEO-Ring), reflecting reduced retrieval performance and increased data gaps over bright surfaces. Combining Terra/Aqua DT and DB (DTB) reduces these discrepancies, but values remain $\sim 18\%$ higher than GEO-Ring. In contrast, single-platform DTB retrievals from Terra or Aqua show even larger uncertainties, averaging 19% and 18% above GEO-Ring, respectively, highlighting the impact of sparse single-overpass temporal sampling. Terra MISR, however, shows closer agreement with GEO-Ring ($\sim 10\%$ higher), but is limited by a low revisit frequency (~ 9 days) and sparse spatial coverage ($\sim 4\%$) (Fig. S16).

The corresponding DARE estimates further reflect these sampling differences. Relative to GEO-Ring, polar-orbiting AOD-derived DARE tends to be more negative over $70\text{--}82\%$ of global land (Fig. 4f–i), indicating an overestimated magnitude of aerosol cooling in these products. Among them, Terra/Aqua MODIS DT yields the most negative annual mean DARE ($-6.8 \pm 4.4 \text{ W m}^{-2}$), $\sim 79\%$ more negative than GEO-Ring (Fig. 4j), likely due to great challenges in retrieving AOD over high-reflectance surfaces (e.g., deserts) and the associated weak surface–atmosphere contrast. Terra/Aqua MODIS DB also produces substantially lower DARE estimates, about 57% below GEO-Ring. Combining Terra and Aqua DTB yields estimates comparable to those from individual platforms, with values remaining $42\text{--}47\%$ lower than GEO-Ring. Large discrepancies are also evident for Terra MISR, which exhibits DARE values that are $\sim 35\%$ less negative than GEO-Ring.

Ground-based measurements provide an independent benchmark to evaluate satellite retrievals. At collocated sites, GEO-Ring shows closer agreement with in situ observations: mean AOD differs by only 0.002 (0.166 ± 0.12 from in situ vs. 0.168 ± 0.12 from GEO-Ring), and mean DARE differs by just 0.3 W m^{-2} ($-4.9 \pm 4.9 \text{ W m}^{-2}$ from in situ vs. $-5.2 \pm 4.2 \text{ W m}^{-2}$ from GEO-Ring), based on calculations using identical aerosol properties (e.g., SSA, AF, and surface albedo), with AOD as the only varying input (Fig. S17). In contrast, polar-orbiting satellites exhibit substantially larger discrepancies relative to ground-based measurements, with AOD biases ranging from -0.01 to 0.05 and negative DARE biases of 0.96 to 2.32 W m^{-2} more negative (Fig. S18). These differences reflect the inability of polar-orbiting satellites to capture the full diurnal cycle of aerosol and cloud variability.

Our AOD and DARE estimates from polar-orbiting satellites are broadly consistent with previous studies (Table S1). However, these earlier studies, like our polar-orbiting results, generally report a stronger aerosol cooling effect than our GEO-Ring estimates. This discrepancy is supported by previous research documenting large diurnal variability in aerosols and its significant impact on radiative effects^{28,44,45}. We show that explicitly accounting for diurnal aerosol and cloud variability using GEO-Ring observations substantially improves global AOD and DARE estimates over land, reducing uncertainties by 7–48% and 35–79%, respectively. These advances, driven primarily by enhanced temporal sampling, with additional contributions from retrieval differences, enable more accurate assessments of climate impacts, air quality, and broader environmental processes.

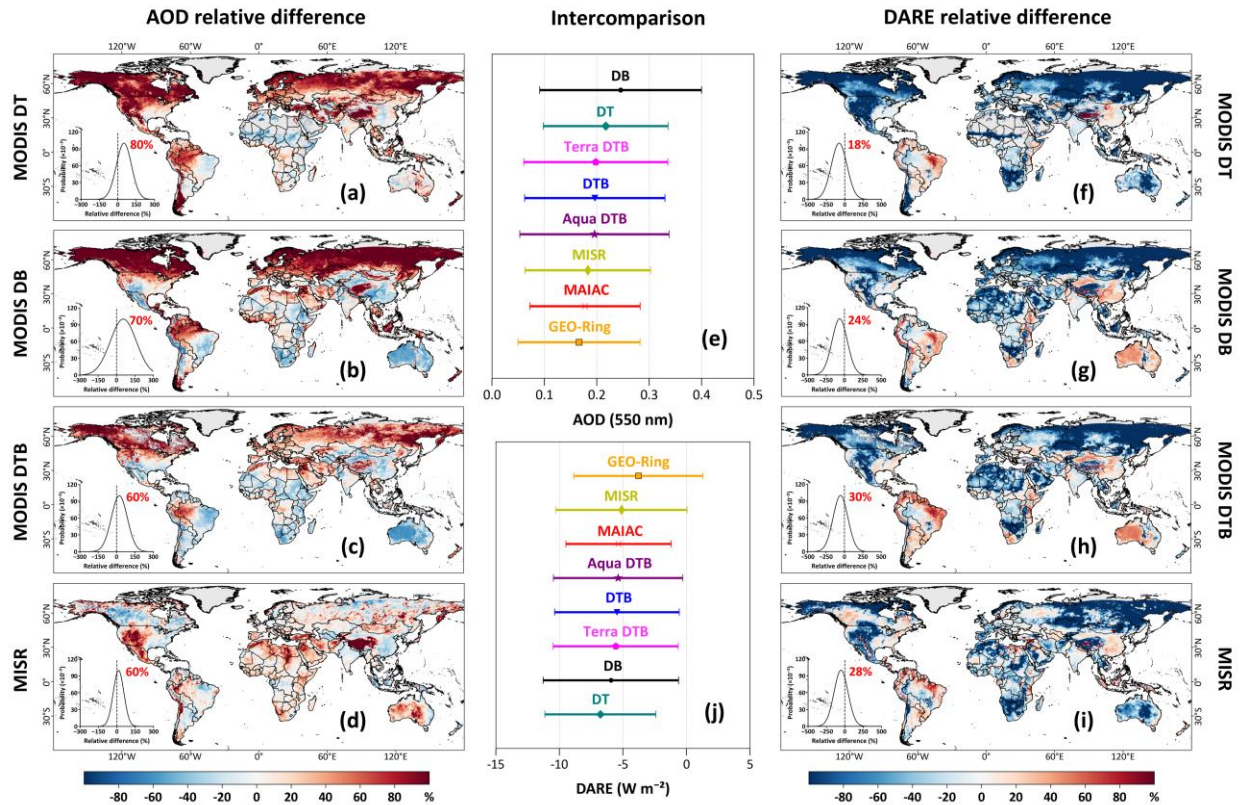


Fig. 4 | Intercomparisons of global AOD and direct radiative effects (DARE) from GEO-Ring and polar-orbiting satellite observations.

Spatial distributions of relative differences in global mean (a–d) AOD and (f–i) all-sky aerosol radiative effect (DARE) between GEO-Ring and different satellite retrieval products and platforms during 2021–2023, including MODIS Dark Target (DT), Deep Blue (DB), combined DT and DB (DTB), and Terra MISR. Panels (e) and (j) show the global mean values and standard deviations of AOD and DARE estimated from each source.

Materials and Methods

GEO-Ring satellite observations

Geostationary (GEO) satellites, orbiting at ~35,786 km above the equator, rotate synchronously with Earth, providing near-continuous observations⁴⁶. In this study, we construct a near-global geostationary aerosol observing system, GEO-Ring, by integrating eight operational platforms: GOES-16/ABI (centered on 75.2°W, launched in 2016), GOES-17/ABI (137.2°W, launched in 2018), and GOES-18/ABI (137.2°W, launched in 2022) over the Americas; MSG-1/SEVIRI (41.5°E, launched in 2002) and MSG-2/SEVIRI (45.5°E, launched in 2005) over Europe, Africa, and western Asia; and FY-4A/AGRI (104.7°E, launched in 2016), Himawari-8/AHI (140.7°E, launched in 2014), and Himawari-9/AHI (140.7°E, launched in 2016) over Asia–Oceania. Together, these satellites provide sub-hourly daytime full-disk imagery covering 92% of global land (Fig. 1). All instruments sample the visible to short-wave infrared (SWIR) range (0.4–2.2 μm), with spatial resolutions of 0.5–1 km in visible bands and approximately 2 km in SWIR bands, and a temporal resolution of 10–15 minutes (Table S2). In this study, to obtain climatologically representative results, we collected all available observations from the eight GEO satellites over the period 2021–2023, applied cloud, snow, and ice screening using the Nowcasting and Very Short Range Forecasting Satellite Application Facility (SAFNWC) masks, and resampled the valid data onto a uniform 2 km $\times$ 2 km grid.

Polar-orbiting satellite aerosol products

Satellite aerosol products from widely used polar-orbiting satellites are employed to compare with our hourly GEO-Ring retrievals, assessing differences in aerosol loadings and associated radiative effects. Four MODIS products are considered: the Multi-Angle Implementation of Atmospheric Correction (MAIAC) AOD (550 nm) product at 1 km resolution as the baseline⁴⁷⁻⁴⁹, Dark Target (DT) and Deep Blue (DB) AOD (550 nm) products at 10 km resolution⁵⁰⁻⁵², and the combined DT–DB (DTB) AOD (550 nm) product at 10 km resolution, which improves both data quality and spatial coverage⁵¹. For each product, instantaneous AOD values from Terra (morning overpass, ~10:30 LT) and Aqua (afternoon overpass, ~13:30 LT) are collected, and daily AOD is obtained by averaging the two overpasses. In addition, the Terra MISR AOD (550 nm) product at 4 km spatial resolution is also included.

MERRA2 reanalysis and auxiliary data

Hourly aerosol diagnostics from MERRA-2 (Modern-Era Retrospective analysis for Research and Applications, Version 2)⁵³, including AOD (550 nm), aerosol scattering AOD (550 nm), and the Ångström exponent (AE; 470–870 nm) at $0.5^\circ \times 0.65^\circ$ resolution, are collected as auxiliary inputs for model development. Hourly total ozone (TOZ) and total precipitable water (TPW) data from MERRA-2 reanalysis are also incorporated to account for ozone and water vapor on absorption effects in satellite-observed multispectral radiances^{54,55}. A 90-m Digital Elevation Model (DEM) from the Shuttle Radar Topography Mission (SRTM) is included to account for topographic effects on aerosol retrieval. All auxiliary variables are resampled to a 2-km resolution using bilinear interpolation to align with GEO-Ring observations.

To estimate the direct aerosol radiative effect, two key aerosol optical properties, including single-scattering albedo (SSA) and asymmetry factor (AF) at 550 nm, are obtained from the monthly Max-Planck Aerosol climatology (MACv3) product. These parameters, which characterize aerosol absorption and scattering, are derived from over two decades of ground-based observations and supplemented with global bottom-up modeling to produce spatially and temporally complete fields⁵⁶. In addition, monthly surface albedo (SA) from the CERES Synoptic TOA and Surface Fluxes and Clouds (SYN1deg Level 3) product is also incorporated²⁶. Here, during the DARE calculation for different satellite retrievals, except for AOD, we use the same surface and aerosol properties (same for SSA, AF, and albedo) to ensure the differences are attributed entirely to AOD. Table S3 provides a detailed summary of the datasets used in this study.

Ground-based aerosol measurements

Aerosol RObotic NETwork (AERONET) is a globally distributed ground-based network of Sun-sky photometers that provides long-term, high-frequency (every 15 min) measurements of multi-wavelength aerosol optical properties, including AOD, under cloud-free conditions⁵⁷. In this study, data from over 470 active stations across more than 200 countries were included, encompassing diverse aerosol regimes such as urban, rural, and desert environments (Fig. 1). AERONET Version 3 AOD observations follow standardized protocols with automatic calibration and robust quality-assurance (Level 2.0), yielding an average uncertainty of $\pm 0.01\%$ ⁵⁷. To enhance spatial coverage in China, where AERONET sites are sparse, AOD measurements from the Sun-sky radiometer

Observation NETwork (SONET) at 15 sites are incorporated (Fig. 1), with an average uncertainty of $\pm 0.5\%$ ⁵⁸. For independent validation, AOD measurements from 27 SKY Radiometer NETwork (SKYNET) stations are also used, these follow the European SKYNET Radiometers (ESR) protocol and show minimal deviation ($\pm 0.3\%$) from AERONET⁵⁹. All AOD measurements are interpolated to 550 nm using the Ångström exponent method³², and measurements within ± 30 min are averaged to match GEO-Ring sampling at each station. In addition, AERONET-derived instantaneous broadband (0.2–4.0 μm) aerosol radiative effects at the top of the atmosphere from the Level 1.5 inversion dataset are used to validate our estimated aerosol radiative effects over land.

AeroTrans-GEO AOD retrieval framework

In this study, we propose AeroTrans-GEO, a globally scalable aerosol retrieval framework that integrates Transformer-informed deep learning with transfer learning to derive hourly AOD over land from a global constellation of geostationary satellites (GEO-Ring). The framework builds upon our previously developed deep learning models for Landsat and MSG/SEVIRI imagery^{32,38}, but incorporates substantial extensions and methodological advances specifically designed to address fundamental limitations of existing geostationary aerosol retrieval methods. These include limited regional transferability, sensitivity to sensor-specific configurations, and the lack of physically consistent AOD retrievals across multiple geostationary platforms.

At the core of AeroTrans-GEO is a Transformer architecture⁶⁰, whose self-attention mechanism enables the model to learn complex, nonlinear aerosol–radiation interactions from high-dimensional, multi-modal inputs while dynamically adapting to varying observation conditions. Unlike conventional radiative-transfer-based algorithms that rely on regionally tuned assumptions or static aerosol models, and unlike convolutional or shallow machine learning approaches that are constrained by local receptive fields, the Transformer explicitly captures long-range dependencies across spectral, geometric, and atmospheric features. This capability is critical for global aerosol retrieval, where aerosol properties, surface reflectance, and atmospheric states vary substantially across climate zones, surface types, and emission regimes³², thereby enhancing the consistency of aerosol retrievals across GEO-Ring. Moreover, the model employs advanced transfer learning with

pretraining to improve spatial robustness and reduce the heavy dependence of traditional machine learning approaches on ground-based stations.

The framework fully exploits the high temporal resolution and quasi-continuous sampling of geostationary satellites, enabling robust characterization of diurnal aerosol cycles and short-lived pollution episodes that are systematically under-sampled by polar-orbiting sensors. By integrating 24-hour observational context through self-attention, the model accurately extracts aerosol information under relatively stable land surface conditions, effectively resolving rapid aerosol evolution associated with anthropogenic emissions, dust mobilization, and biomass burning, thereby providing a more complete representation of aerosol radiative effects.

To ensure global adaptability and cross-platform consistency, AeroTrans-GEO employs a two-stage transfer-learning strategy consisting of large-scale pre-training followed by targeted fine-tuning. During the pre-training stage, a unified Transformer (TF) model is trained using physically interpretable, sensor-agnostic inputs, including multi-channel top-of-atmosphere reflectance (TOA_λ), observation geometry (solar zenith angle, SZA ; view zenith angle, VZA ; and scatter angle, SCA), atmospheric composition (TOZ and TPW), surface elevation (DEM), and aerosol priors from the MERRA-2 reanalysis (hourly AE , SSA , and daily AOD). These inputs are deliberately selected to maximize physical consistency across geostationary platforms while minimizing sensitivity to platform-specific radiometric and spectral differences, thereby providing physically informed initial constraints on aerosol properties. Hourly MERRA-2 AOD ($MAOD$) is used as the training target, allowing the model to learn globally coherent aerosol–radiative relationships under a wide range of atmospheric and surface conditions.

The pre-training dataset is constructed by collocating geostationary observations with MERRA-2 aerosol fields, yielding a large, diverse sample set with approximately two million records subsampled from ~10 million candidate matches. This dataset spans near all land-cover types, including grassland (24%), barren (16%), forest (15%), shrubland (13%), savannas (11%), cropland (10%), and urban (2%), and various atmospheric conditions. This provides the model with strong a priori knowledge before exposure to ground-based observations.

In the fine-tuning stage, the pre-trained Transformer model is adapted to real-world conditions by updating the final two hidden layers using the same multi-modal input feature set $X(all)$, while replacing the training target with ground-based hourly AOD measurements ($GAOD$) from the AERONET and SONET networks as the target.

$$\frac{MAOD}{GAOD} \leftrightarrow \frac{TF_{Pre-train}}{TF_{Fine-tune}} \left[\begin{array}{c} GEO - Ring \left(\begin{array}{c} GOES - 18: TOA_{0.47,0.64,0.86,1.61,2.25}, \\ GOES - 17: TOA_{0.47,0.64,0.86,1.61,2.25}, \\ GOES - 16: TOA_{0.47,0.64,0.86,1.61,2.25}, \\ MSG - 1: TOA_{0.64,0.81,1.64}, \\ MSG - 2: TOA_{0.64,0.81,1.64}, \\ FY - 4A: TOA_{0.47,0.65,0.86,1.38,1.61,2.25}, \\ Himawari - 8: TOA_{0.47,0.51,0.64,0.86,1.61,2.26}, \\ Himawari - 9: TOA_{0.47,0.51,0.64,0.86,1.61,2.26} \end{array} \right) + \\ Auxilliary(SZA, VZA, SCA, TOZ, TPW, DEM, AE, SSA, AOD_{MERRA2}^{daily}) \end{array} \right] \quad (1)$$

Here, $\leftrightarrow$ denotes the iterative pre-training and fine-tune process between model inputs and outputs. All input features are normalized using Min–Max scaling prior to training, and the fitted scaler is retained to ensure consistency and reproducibility during subsequent fine-tuning and inference.

This fine-tuning strategy enables efficient adjustment to sensor-specific characteristics—such as spectral response functions, calibration uncertainties, and viewing geometry—without retraining the entire model from scratch, thereby ensuring retrieval continuity and physical consistency across the GEO-Ring constellation. Transfer learning substantially enhances retrieval robustness in data-sparse regions and improves generalization over complex terrain and persistently cloudy areas, where conventional aerosol retrieval methods often fail.

Further details of the pre-training and fine-tuning procedures are provided in Text S2. Following fine-tuning, the model demonstrates a significantly improved capability for retrieving hourly AOD under real-world conditions, particularly across vast remote regions with sparse ground-based monitoring coverage, consistently outperforming direct, non-transfer-learning baselines. Together, the integration of Transformer-based representation learning with transfer learning, combined with consistent input normalization using preserved Min–Max scalers, establishes a unified, flexible, and globally applicable framework for geostationary aerosol retrieval. This framework enables

physically consistent, high-temporal-resolution AOD products across multiple geostationary platforms within the GEO-Ring constellation.

Direct aerosol radiative effect (DARE) estimation

The direct aerosol radiative effect (DARE; unit: W m^{-2}) is defined as the change in net shortwave irradiance due to aerosols. Here, we focus on clear-sky, top-of-atmosphere shortwave DARE over land within the 0.25–4 μm spectral range:

$$DARE = (F^\downarrow - F^\uparrow)_{with\ aerosol} - (F^\downarrow - F^\uparrow)_{no\ aerosol} \quad (2)$$

DARE is commonly computed using radiative transfer models (RTMs), which require inputs describing solar geometry, aerosol optical properties, and surface reflectance, including aerosol optical depth (AOD), solar zenith angle (SZA), single-scattering albedo (SSA), asymmetry factor (AF), and surface albedo (SA)¹⁷. Due to the limited temporal sampling of polar-orbiting satellites, previous studies have often approximated daily mean DARE by using instantaneous observations as proxies for daytime averages. This approach, however, does not account for the diurnal variability of aerosols and clouds.

In contrast, we use hourly aerosol fields derived from GEO-Ring observations to explicitly resolve diurnal variations. For each hour, DARE is calculated under clear-sky conditions when AOD retrievals are available. To obtain the daily mean DARE, which represents the effective aerosol radiative impact in the presence of clouds, we assume zero aerosol radiative effect during cloudy hours, that is, when AOD retrievals are unavailable due to cloud cover, and average the results over all daytime hours. For each land pixel (i, j) , the daily mean DARE is calculated as:

$$DARE_D^{clear}(i, j) = \frac{1}{N_{i,j}} \sum_{t \in H_{i,j}} DARE_H(AOD_{i,j}^{clear}, SZA_{i,j}^{clear}, SSA_{i,j}^{clear}, AF_{i,j}^{clear}, SA_{i,j}^{clear}) \quad (3)$$

where, $H_{i,j}$ denotes the set of hourly timestamps meeting cloud-free and geometry quality criteria, and $N_{i,j} = |H_{i,j}|$ is the number of valid hours during daytime. To enable efficient continent-to-global calculations, we replace computationally expensive pixel-wise RTM calls with a machine

learning surrogate trained on RTM simulations. Specifically, we generate a synthetic training set using the Santa Barbara DISORT Atmospheric Radiative Transfer (SBDART) model under plane-parallel, clear-sky conditions, with inputs including AOD, SZA, SSA, AF, and SA sampled over the parameter ranges detailed in Table S4 (see additional configuration in Text S3). A lightweight Extremely Randomized Trees (ERT) model⁶¹ was then employed to learn the relationship between these input variables and DARE as simulated by the radiative transfer model:

$$DARE_H \leftrightarrow ERT(SZA, AOD, SSA, AF, SA) \quad (5)$$

Here, “ $\leftrightarrow$ ” indicates the iterative training and mapping process between RTM inputs and surrogate model outputs. Finally, global aerosol radiative effects over land were estimated using the trained ERT model with aerosol and surface inputs, enabling rapid and accurate DARE predictions for all land pixels.

Validation strategies and results

We evaluated the hourly AOD retrievals from the AeroTrans-GEO model using multiple independent validation strategies. First, a random ten-fold cross-validation (CV) was performed, partitioning all samples into ten folds and iteratively training on nine folds (90%) while validating on the remaining fold (10%) to obtain out-of-sample estimates as a baseline. To mitigate biases arising from the uneven global distribution and clustering of ground-based monitoring stations, we further conducted spatially independent validation by partitioning training and validation data across hierarchical geographic scales, from station to state, country, and continent, thereby assessing the model’s extrapolative skill in regions lacking ground-based measurements. Temporal generalization was evaluated via ten-fold CV with progressively larger holdout intervals, including hours, days of the year, weeks, and months. Additionally, retrievals were validated against SKYNET AOD measurements not used in model training, offering a fully independent external assessment of model performance.

To isolate the effect of diurnal sampling on AOD validation, we designed three matchup strategies with progressively relaxed temporal constraints. First, GEO-Ring AOD was paired with AERONET observations at all available times to establish a benchmark for full-diurnal agreement.

Second, the matchup was restricted to the Terra (10:30 local time) and Aqua (13:30 local time) overpass times to emulate polar-orbiting sampling. Finally, GEO-Ring observations sampled only at these overpass times were compared with daily AERONET averages to quantify the bias introduced by fixed-time sampling.

For DARE estimates, ten-fold CV was applied to assess the ERT surrogate model using simulated samples from SBDART. Additionally, considering the influence of surface albedo on DARE, we performed a surface-albedo-stratified ten-fold CV, partitioning samples by surface albedo values to evaluate the model's predictive skill under unseen surface conditions. Finally, DARE estimates were validated against AERONET measurements to provide an independent evaluation of model performance under real-world conditions (Text S4).

Data availability

The GEO-Ring dataset will be released after our forthcoming study; details on the original data availability is provided in Supplement Materials (Text S5).

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